Launch Linux EC2 instances in two regions using a single Terraform file.

We will create one Terraform file that launches EC2 in:

- **us-east-1**
- **us-west-2**

1️⃣ Create a Project Folder

mkdir terraform-ec2

cd terraform-ec2

```
Complete!
~ $ terraform -version
Terraform v1.14.6
on linux_amd64
~ $ mkdir terraform-ec2
~ $ cd terraform-ec2
terraform-ec2 $
```

2️⃣ Create Terraform File

nano main.tf (Code added in the readme file)

```
      + primary_network_interface (known after apply)
  GNU nano 8.3
provider "aws" {
  alias  = "east"
  region = "us-east-1"
}

provider "aws" {
  alias  = "west"
  region = "us-west-2"
}

# Get latest Amazon Linux AMI for us-east-1
data "aws_ami" "amazon_linux_east" {
  provider    = aws.east
  most_recent = true

  owners = ["amazon"]

  filter {
    name   = "name"
    values = ["al2023-ami-*-x86_64"]
  }
}

# Get latest Amazon Linux AMI for us-west-2
data "aws_ami" "amazon_linux_west" {
  provider    = aws.west
  most_recent = true

  owners = ["amazon"]

  filter {
    name   = "name"
```

3️⃣ Initialize Terraform

terraform init

4️⃣ **Validate the Code**

terraform validate

```
Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
terraform-ec2 $ terraform validate
Success! The configuration is valid.

terraform-ec2 $
```

5️⃣ **Preview Infrastructure**

terraform plan

6️⃣ **Launch EC2 Instances**

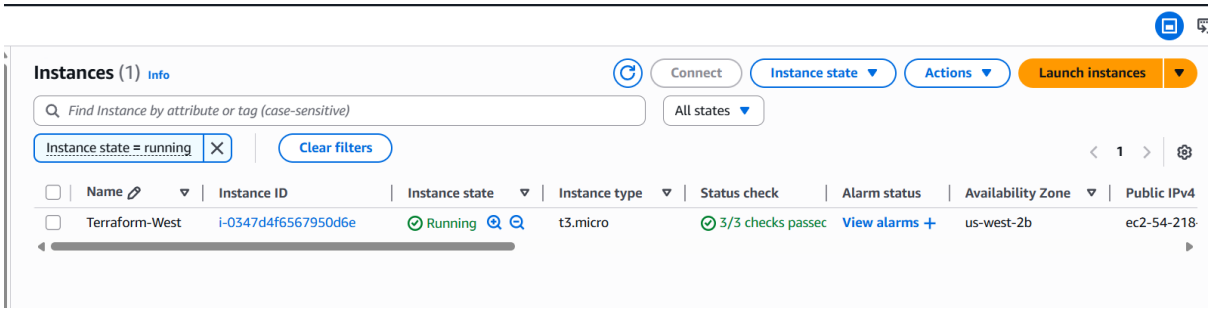
terraform apply

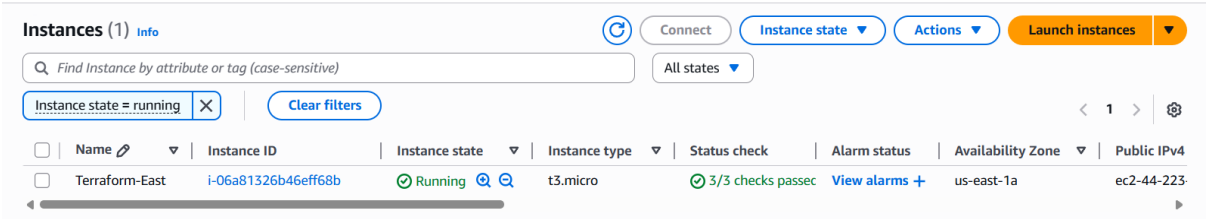
```
aws_instance.ec2_east: Destroying... [id=i-0e1492eaf9e03d349]
aws_instance.ec2_west: Creating...
aws_instance.ec2_east: Still destroying... [id=i-0e1492eaf9e03d349, 00m10s elapsed]
aws_instance.ec2_west: Still creating... [00m10s elapsed]
aws_instance.ec2_west: Creation complete after 14s [id=i-0347d4f6567950d6e]
aws_instance.ec2_east: Still destroying... [id=i-0e1492eaf9e03d349, 00m20s elapsed]
aws_instance.ec2_east: Still destroying... [id=i-0e1492eaf9e03d349, 00m30s elapsed]
aws_instance.ec2_east: Still destroying... [id=i-0e1492eaf9e03d349, 00m40s elapsed]
aws_instance.ec2_east: Still destroying... [id=i-0e1492eaf9e03d349, 00m50s elapsed]
aws_instance.ec2_east: Still destroying... [id=i-0e1492eaf9e03d349, 01m00s elapsed]
aws_instance.ec2_east: Still destroying... [id=i-0e1492eaf9e03d349, 01m10s elapsed]
aws_instance.ec2_east: Destruction complete after 1m10s
aws_instance.ec2_east: Creating...
aws_instance.ec2_east: Still creating... [00m10s elapsed]
aws_instance.ec2_east: Creation complete after 13s [id=i-06a81326b46eff68b]

Apply complete! Resources: 2 added, 0 changed, 1 destroyed.
```

Now verify in AWS Console

Go to **EC2 Dashboard** and check in these regions:

Region	Instance Name
us-east-1	Terraform-East
us-west-2	Terraform-West


Instances (1) Info
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Instance state = running
Clear filters
1
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
Terraform-West
i-0347d4f6567950d6e
Running
t3.micro
3/3 checks passed
View alarms +
us-west-2b
ec2-54-218


Instances (1) Info
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Instance state = running
Clear filters
1
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
Terraform-East
i-06a81326b46eff68b
Running
t3.micro
3/3 checks passed
View alarms +
us-east-1a
ec2-44-223